

1. Cloud Network Bandwidth Pricing

Background:

In a **cloud network**, an infinite number of **data centers** are set up globally. Each data center is assigned a unique positive integer starting from 1. The cloud network has direct, bidirectional data channels between:

1. Data center i and data center $2i$.
2. Data center i and data center $2i + 1$.

There exists a unique shortest path between any two data centers.

Initially, data transfer between any two data centers is free. However, due to network optimization, the cloud provider occasionally imposes bandwidth fees on certain data channels. The network provider will make several updates to the fees based on upcoming events.

There are two types of events:

1. Fee Update: The cloud provider updates the bandwidth fee for all data channels along the shortest path between data centers a and b by adding a fee of x units.

2. Data Transfer: A customer needs to transfer data from one data center a to another data center b . The transfer always follows the shortest path. You need to calculate the total cost incurred for this data transfer.

Input:

- The first line contains an integer q ($1 \leq q \leq 1,000$), representing the number of events.
- The next q lines contain details of the events in chronological order:
- $1 \ a \ b \ x$ indicates a fee update where all channels on the shortest path from a to b increase their fee by x units.
- $2 \ a \ b$ represents a data transfer request from data center a to data center b .
- They are present in the form of 2D array of q rows and 4 columns.

Output:

For each data transfer event ($2 \ a \ b \ 0$), add to the sum the total cost for the transfer and return the sum.

Function description

Complete the **solve** function in the editor below. It has the following parameter(s):

Name	Type	Description
q	INTEGER	number of events
a	INTEGER 2D ARRAY	details of the events in chronological order

Return

The function must return an **INTEGER** denoting the sum of total cost for each transfer event.

Constraints

- $1 \leq q \leq 10^3$
- $1 \leq a[i][j] \leq 10^9$